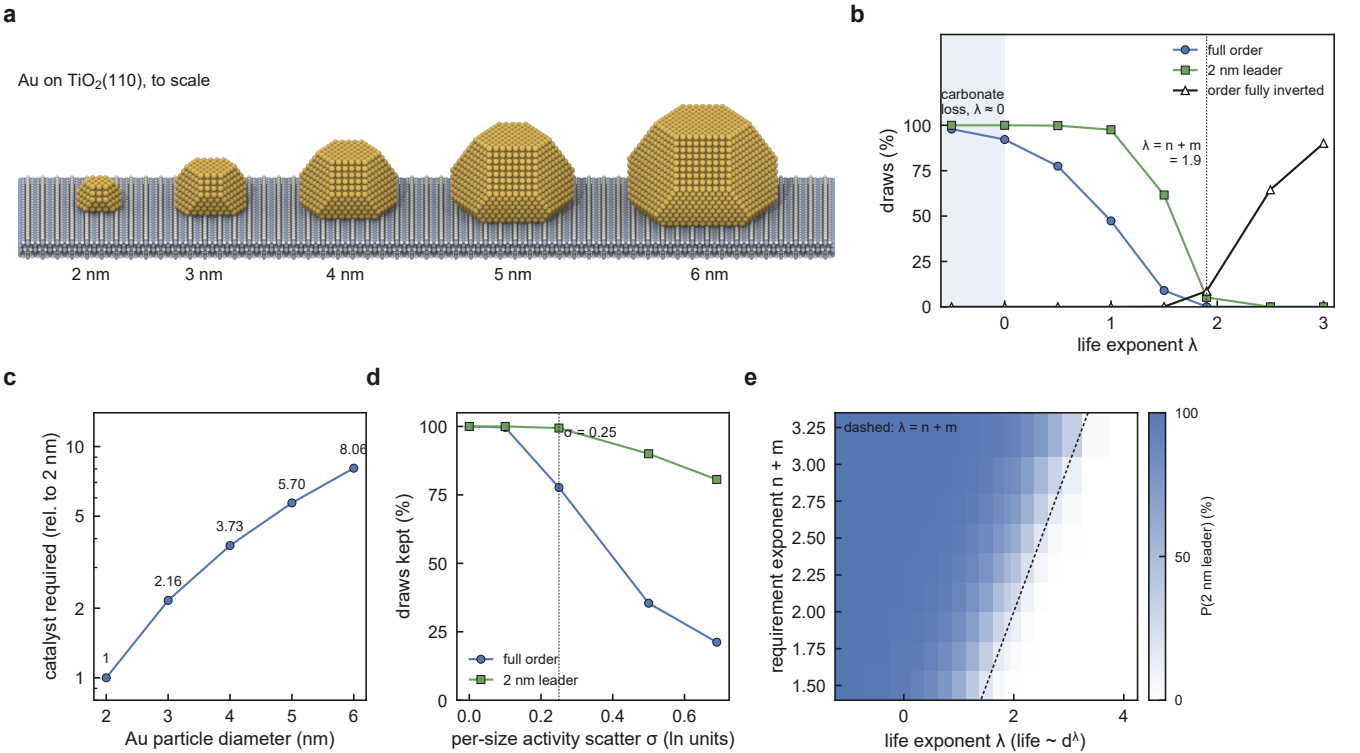

a
Au on TiO2(110), to scale
2 nm
3 nm
4 nm
5 nm
6 nm
b
full order
2 nm leader
order fully inverted
carbonate loss, λ ≈ 0
λ = n + m = 1.9
draws (%)
life exponent λ
c
catalyst required (rel. to 2 nm)
Au particle diameter (nm)
1
2.16
3.73
5.70
8.06
d
draws kept (%)
per-size activity scatter σ (ln units)
σ = 0.25
full order
2 nm leader
e
dashed: λ = n + m
requirement exponent n + m
life exponent λ (life ~ d^λ)
P(2 nm leader) (%)